

● MEDIUM

● ALGEBRA

Algebra

10 questions · ~15 min

18

Q1

ALGEBRA

$$y = 3x + 9$$

$$3y = 8x - 6$$

The solution to the given system of equations is x, y . What is the value of $x - y$?

- A. -123
- B. -33
- C. 3
- D. 57

Line p is defined by $2y + 18x = 9$. Line r is perpendicular to line p in the xy -plane. What is the slope of line r ?

A. -9

B. $-\frac{1}{9}$

C. $\frac{1}{9}$

D. 9

The table gives the number of hours, h , of labor and a plumber's total charge fh , in dollars, for two different jobs.

h	fh
1	155
3	285

There is a linear relationship between h and fh . Which equation represents this relationship?

- A. $fh = 25h + 130$
- B. $fh = 130h + 25$
- C. $fh = 65h + 90$
- D. $fh = 90h + 65$

$$x + 3 = -2y + 5$$

$$x - 3 = 2y + 7$$

The solution to the given system of equations is x, y . What is the value of $2x$?

- A. -2
- B. 6
- C. 12
- D. 24

The pressure exerted on a scuba diver at sea level is 14.70 pounds per square inch psi. For each foot the scuba diver descends below sea level, the pressure exerted on the scuba diver increases by 0.44 psi. What is the total pressure, in psi, exerted on the scuba diver at 105 feet below sea level?

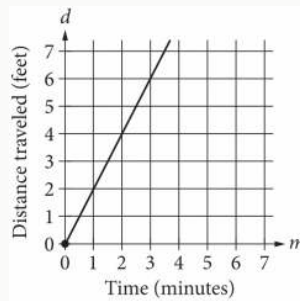
- A. 60.90
- B. 31.50
- C. 14.70
- D. 0.44

$$y = -\frac{1}{5}x$$

$$y = \frac{1}{7}x$$

The solution to the given system of equations is x, y . What is the value of x ?

- A. -5
- B. 0
- C. 2
- D. 7



The graph above shows the distance traveled d , in feet, by a product on a conveyor belt m minutes after the product is placed on the belt. Which of the following equations correctly relates d and m ?

A. $d = 2m$

B. $d = \frac{1}{2}m$

C. $d = m + 2$

D. $d = 2m + 2$

$$f(x) = 2x + 244$$

The given function f represents the perimeter, in centimeters, of a rectangle with a length of x cm and a fixed width. What is the width, in cm, of the rectangle?

- A. 2
- B. 122
- C. 244
- D. 488

Distance (kilometers)	Average time (minutes)
0.32	8
0.56	14
0.68	17

The table gives the average time t , in minutes, it takes Carly to travel a certain distance d , in kilometers. Which equation could represent this linear relationship?

A. $t = 4d$

B. $t = \frac{1}{25}d$

C. $t = 25d$

D. $t = \frac{1}{4}d$

x	gx
1	32
2	28
3	24
4	20

For the linear function g , the table shows four values of x and their corresponding values of gx . The function can be written as $gx = mx + b$, where m and b are constants. What is the value of b ?

- A. 4
- B. 16
- C. 32
- D. 36